

Android Python Automation Assignment

Clarified requirements and submission guide

Build a small command line tool in Python that uses Android Debug Bridge to inspect one Android device or emulator, run five checks, and write a readable test report. The final submission must include working source code, configuration, automated unit tests, and a README.

What You Must Deliver

- A complete Python project that runs from the command line.
- A config.json file containing the package name and test thresholds.
- A report.txt file produced by a normal program run.
- A logs directory used for logcat output when a test fails.
- Automated unit tests that can run without a real Android device by mocking ADB.
- A README with setup, usage, assumptions, failure behavior, and improvement ideas.

Required Program Flow

1. Load and validate config.json before contacting a device.
2. Run adb devices and select one authorized device or emulator.
3. Collect the required device information.
4. Run the five checks in the test matrix below.
5. Capture adb logcat output whenever a check fails.
6. Write report.txt with results, totals, and the overall outcome.
7. Exit cleanly with a useful message for expected setup or ADB failures.

Configuration

Keep user-editable values in JSON rather than embedding them in Python. The three fields below are required. A device_id field may be added to support explicit selection when several devices are connected.

```
{  
  "package_name": "com.example.demo",  
  "minimum_android_version": 10,  
  "minimum_battery": 20  
}
```

Device Detection

Run adb devices and parse each device serial and state. Continue only when a usable device has state device. Report no-device, unauthorized, and offline conditions in plain language. If more than one usable device is connected, either select the configured device_id or ask the user to configure one.

Device Information

Collect and include all five values in the report:

- Device ID or emulator serial
- Android version
- Device model
- Manufacturer
- Battery level as a percentage

Required Tests

Test	Required behavior	Pass condition
Device is reachable	Run an ADB shell command such as echo hello.	The command succeeds and returns the expected response.
Android version	Read the OS version and compare its major version with the configured minimum.	Installed version is equal to or greater than the minimum.
Battery level	Read the current battery percentage.	Battery is strictly above the configured threshold.
Application installation	Check whether the configured package exists on the device.	ADB reports a package path for the exact package name.
Launch application	When installed, launch the package through ADB and inspect the command result.	The launch command succeeds using a documented, reasonable command-level check.

If the application is not installed, report the installation test as FAIL and the launch test as SKIP. A command-level launch check is sufficient; visual UI verification is outside this assignment.

Report Requirements

Write report.txt after running the checks. It must contain:

- The device information listed above.
- Each test name, PASS, FAIL, or SKIP status, and a short explanation.
- Totals for passed, failed, skipped, and all tests.
- An overall PASS when no test failed; otherwise an overall FAIL.

Failure Logs

When a check returns FAIL, run adb logcat -d and save the output under logs. Use a unique, readable filename such as a timestamp plus the test name. Log capture is best effort: if logcat also fails, preserve the original test result and continue writing the report.

Expected Error Handling

Condition	Expected response
ADB is missing	Explain that Android Platform Tools must be installed and adb added to PATH.
No device	State that no Android device or emulator is connected.
Unauthorized or offline	Name the device and its state so the user can correct it.
Invalid configuration	Identify the missing or invalid field without showing a raw traceback.
ADB command fails or times out	Report a concise command error and stop or fail the affected check safely.

Automated Testing

Provide several unit tests for the Python logic. Mock subprocess execution so the tests do not depend on a connected device. At minimum, cover valid and invalid configuration, online and unauthorized device detection, at least one complete passing run, one failing check, log capture, and report totals.

Suggested Project Structure

```
android_automation_solution/  
|-- main.py  
|-- config.json  
|-- android_runner/  
|   |-- adb.py  
|   |-- config.py  
|   |-- runner.py  
|   |-- report.py  
|-- tests/  
|-- README.md  
|-- logs/
```

README Requirements

Explain each of the following in the README:

- Prerequisites and device setup
- How to edit configuration and run the tool
- How to run the unit tests
- Project structure and important design choices
- Expected failure behavior and assumptions
- Improvements you would make with more time

Acceptance Checklist

Check	Acceptance criterion
Execution	The project starts with a documented Python command.
Configuration	Invalid JSON or values produce a clear message.
Device handling	No-device, unauthorized, offline, and multiple-device states are handled.
Test coverage	All five required checks produce explicit results.
Reporting	The report includes device details, each result, totals, and overall status.
Logs	Each failed check attempts to save logcat output under logs.
Unit tests	Tests run without a real device and cover success and failure paths.
Documentation	The README gives another developer enough information to run and assess the project.

Optional Extensions

- Select a specific device when several are connected.
- Run continuously at a configurable interval.
- Filter or analyze logcat output.
- Generate JSON or HTML in addition to the text report.
- Add further checks such as storage, network, foreground activity, or screen state.

Submission

Submit the complete project directory. Include source code, config.json, automated tests, README.md, and any useful sample output. Do not submit credentials, private device data, or unrelated build files.